

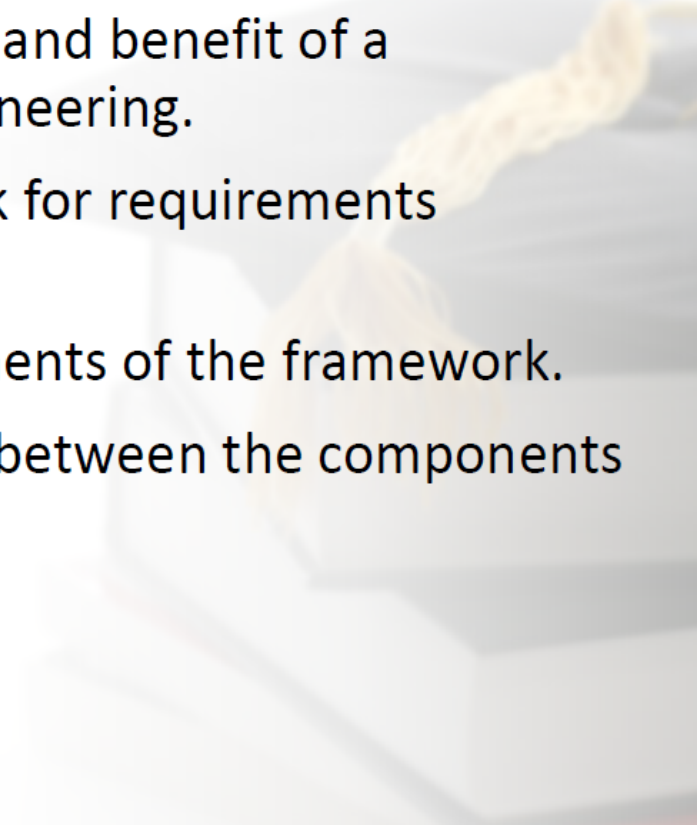
CKS121/CSE 242 Software Requirements

Chapter 2: Requirement Engineering (RE) Framework

Learning Outcomes

The students

- understand the motivation, goals and benefit of a framework for requirements engineering.
- get an overview of the framework for requirements engineering.
- get a first insight into the components of the framework.
- understand the key relationships between the components of the framework.



Structure of this Lecture

- **Motivation, Goals & Overview**
- Requirements Engineering Context
- Core Activities
- Requirements Artefacts
- Cross-Sectional Activities

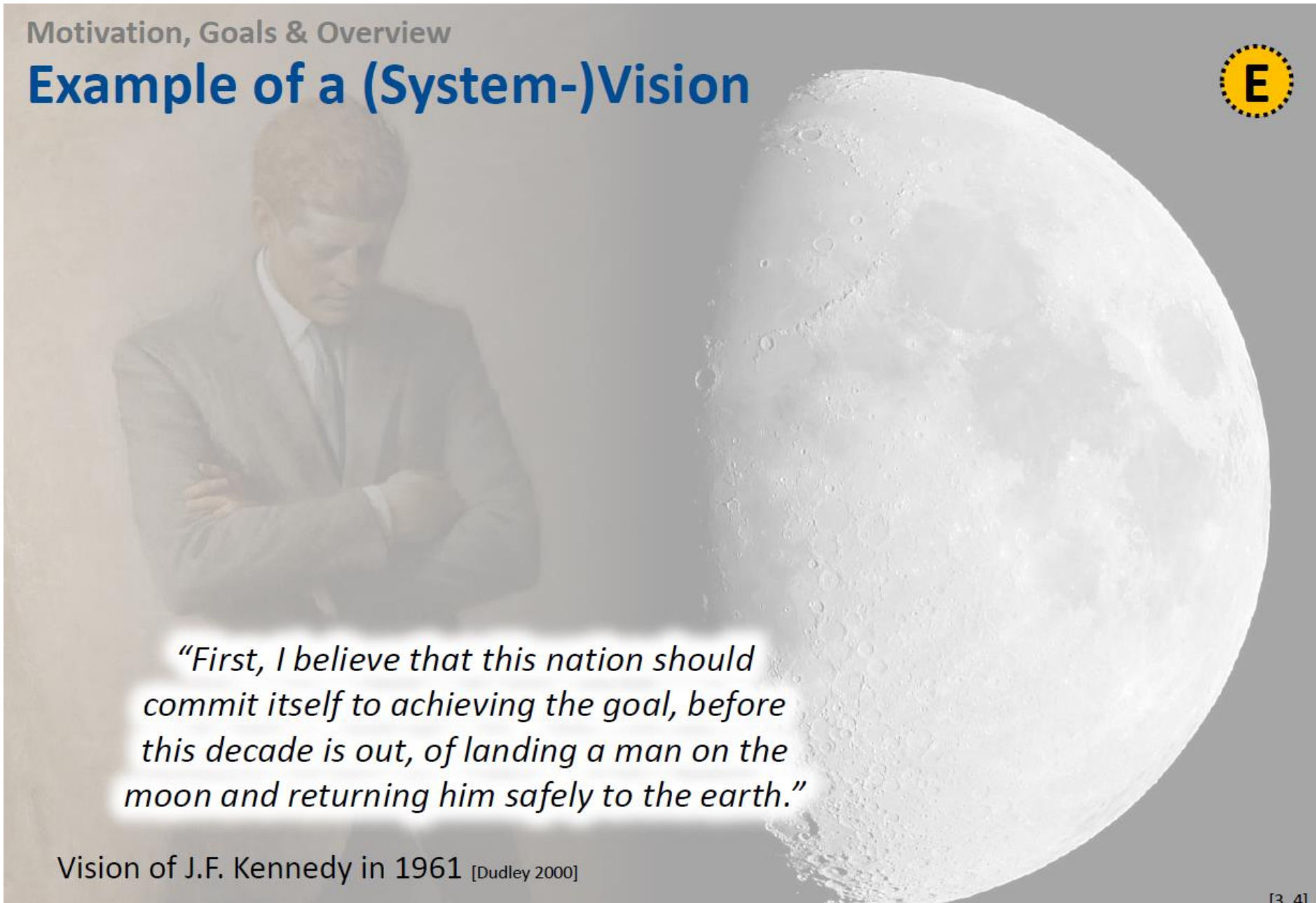


Starting the Requirements Engineering Process

Idea/wish to change
the current reality!



Example of a (System-)Vision



“First, I believe that this nation should commit itself to achieving the goal, before this decade is out, of landing a man on the moon and returning him safely to the earth.”

Vision of J.F. Kennedy in 1961 [Dudley 2000]

Example of a (System-)Vision



Small changes to the vision
might have huge impact on the
requirements and the way the
system is being build.

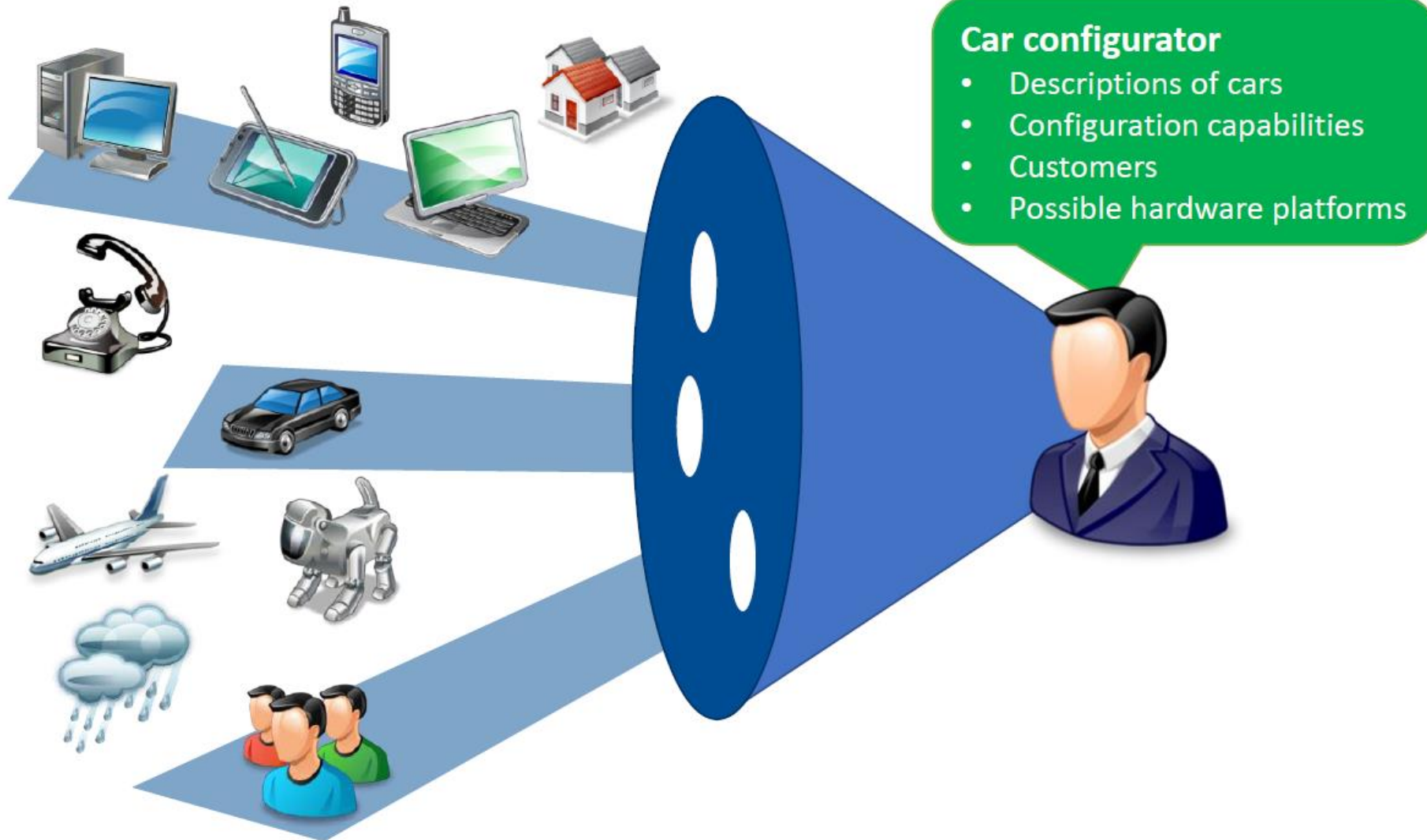
*“First, I believe that this nation should
commit itself to achieving the goal, before
this decade is out, of landing a man on the
moon ~~and returning him safely to the earth.~~”*

Vision of J.F. Kennedy in 1961 [Dudley 2000]

Vision

- Defines an intended (small or large) change to a current reality (differs from reality).
- Is typically brief and precise.
- Guides the definition of requirements.
- Guides the development of the system.
- States a goal (“What?”),
not how to achieve it (“How?”).
- Is the basis for making decisions.
- Justifies expenses.

Vision as Focus



Establishing a Vision in Context (1)

- Each system is embedded in a specific context.
- The context contains, among others, the sources for the requirements.
- The context strongly influences the definition and refinement of requirements.
- Adequate consideration of the context is thus essential for requirements engineering and system development.
- The context is often not fully known and understood.

Establishing a Vision in Context (2)



Vision: “The system shall stop from a speed of 50 km/h within 10 meters.”



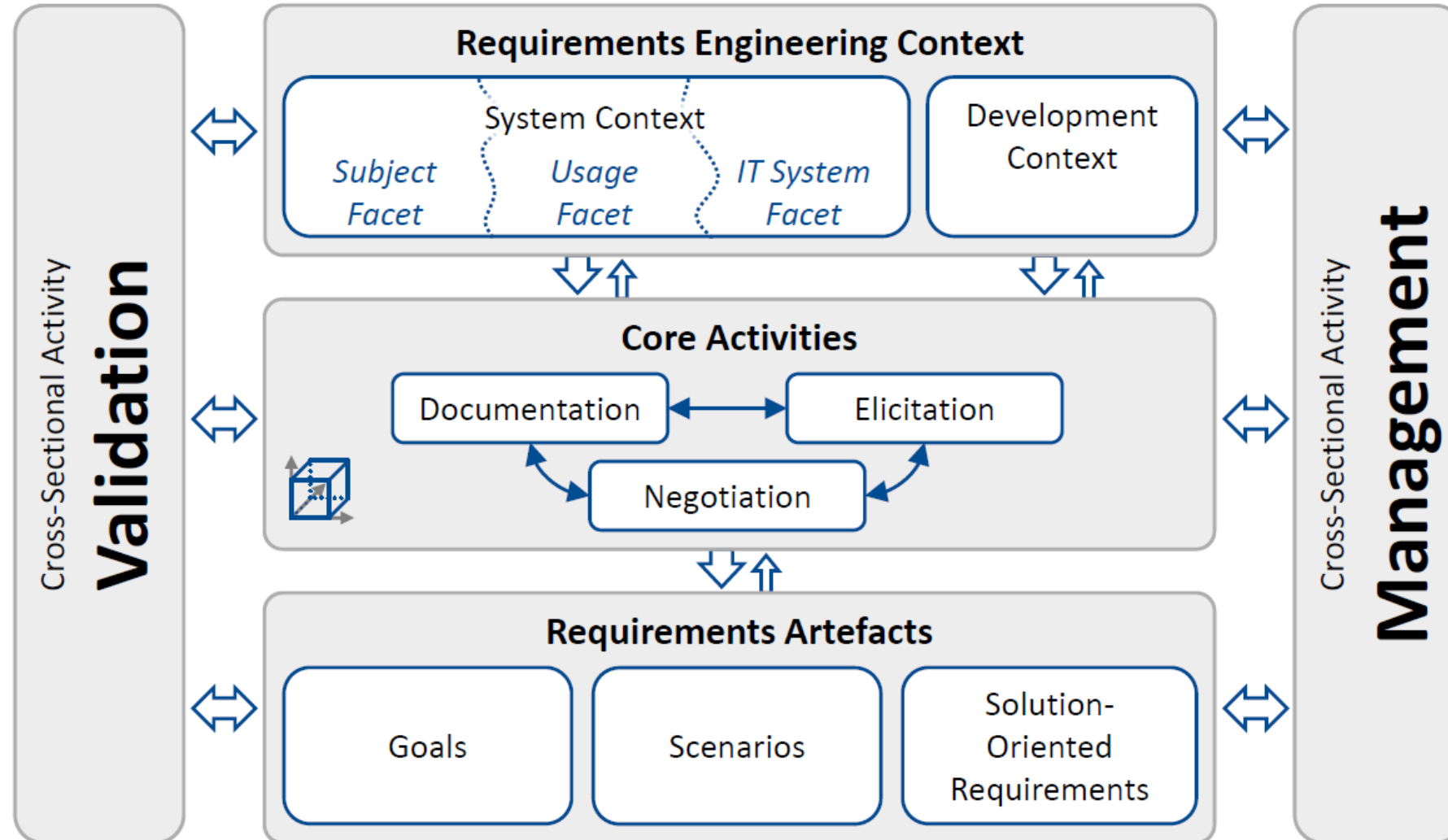
The **realization** of this “vision” **differs largely** in each of the above **contexts**.

In some contexts the vision can hardly be realised, e.g., a train or a ship.

Motivation for the Framework

- Structuring requirements engineering by defining a set of core common concepts for each requirements engineering process and their relationships.
- Reference structure for industry
 - Training of managers, requirements engineers and developers.
 - Analysis of strength and weaknesses of RE processes.
- Reference structure for teaching requirements engineering.
- Successfully introduced in a number of organizations, companies and universities!

Framework for Requirements Engineering

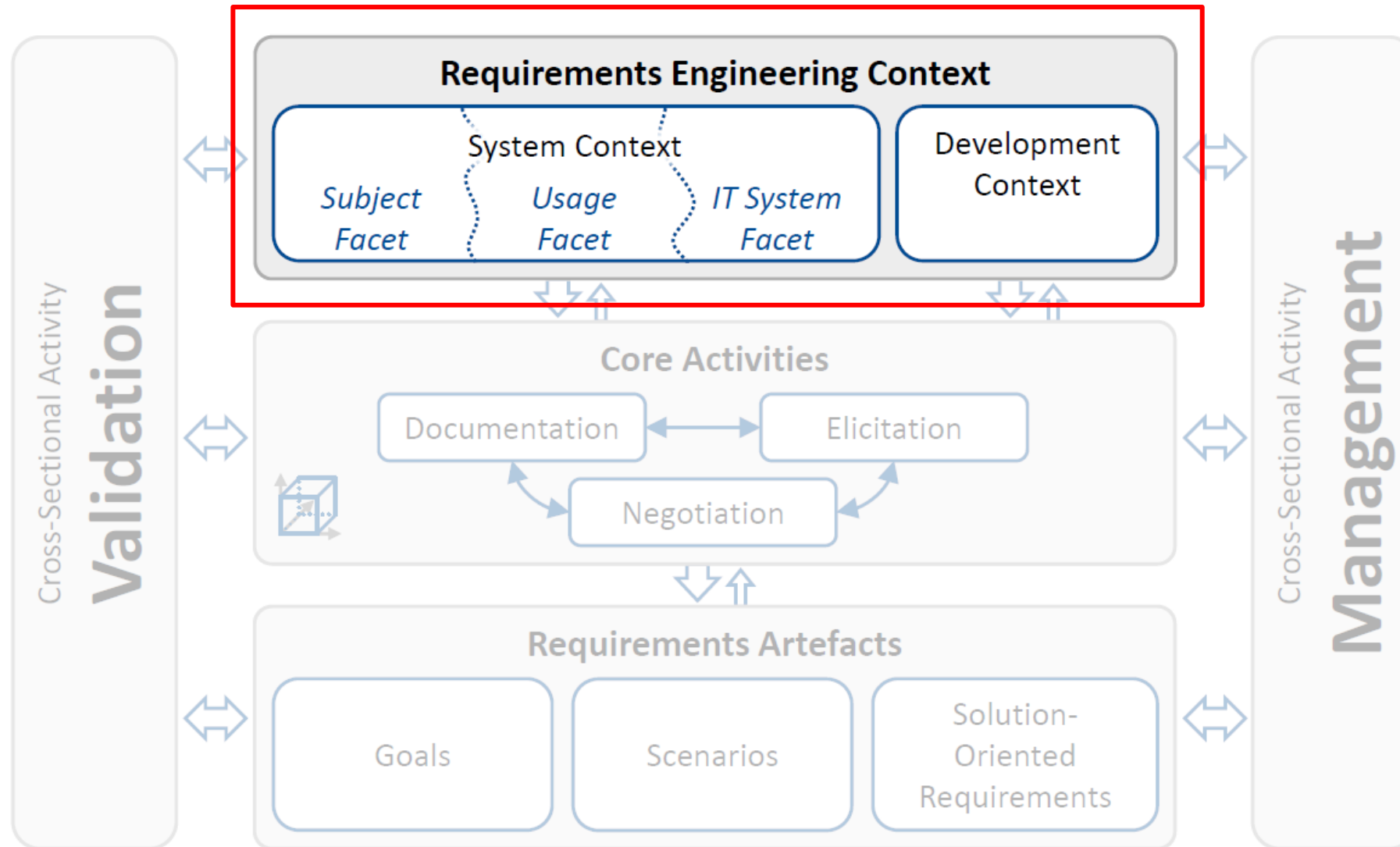


A background image showing a hand holding a pencil, poised to draw on a map. The map features various geographical outlines and labels, though they are not clearly legible. The overall color palette is warm, with shades of orange, yellow, and brown.

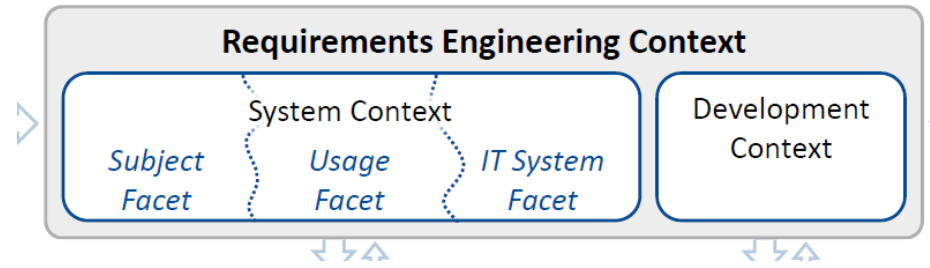
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Framework for Requirements Engineering



Constituent Parts of the Context

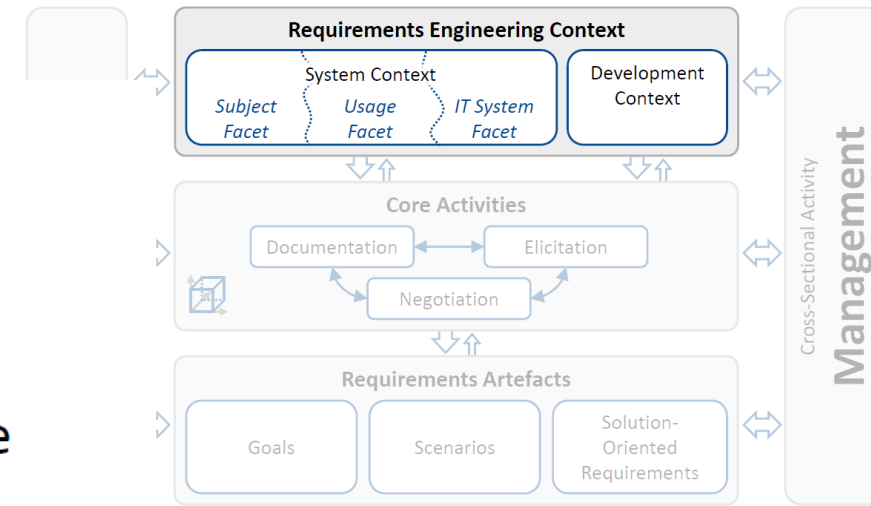


The requirements engineering context consists of

- (1) **The system context** is the part of the context in which the system to be developed is operating/ embedded.
- (2) **The development context** is the part of the context in which the system is being developed.
- (3) **Additional requirements engineering context objects** which have to be considered during requirements engineering, but which are not part of the system context nor the development context.

Three Facets of the System Context

- **Subject** facet
comprises system context objects about which information is represented in the system or which influence or constrain the representation of information represented in the system.
- **Usage** facet
comprises all system context objects (people and/or systems) which directly or indirectly interact with the system or which influence or benefit from the usage of the system.
- **IT system** facet
comprises all system context objects of the technical and operational environment in which the system is going to be deployed in or which influence or constraint the deployment of the system and/or the use of technology by the system such as sensors, actuators or other systems.

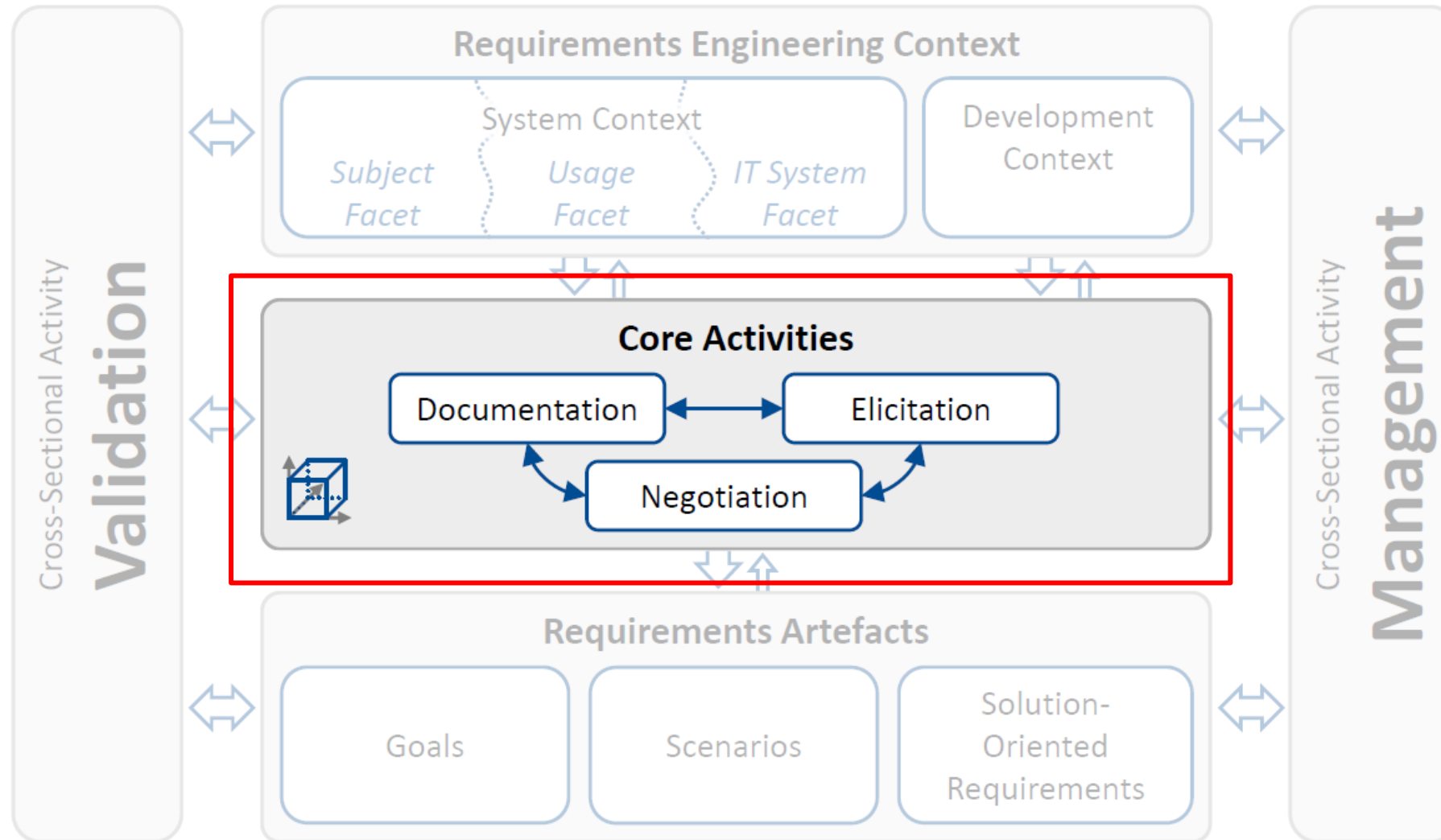


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Framework for Requirements Engineering

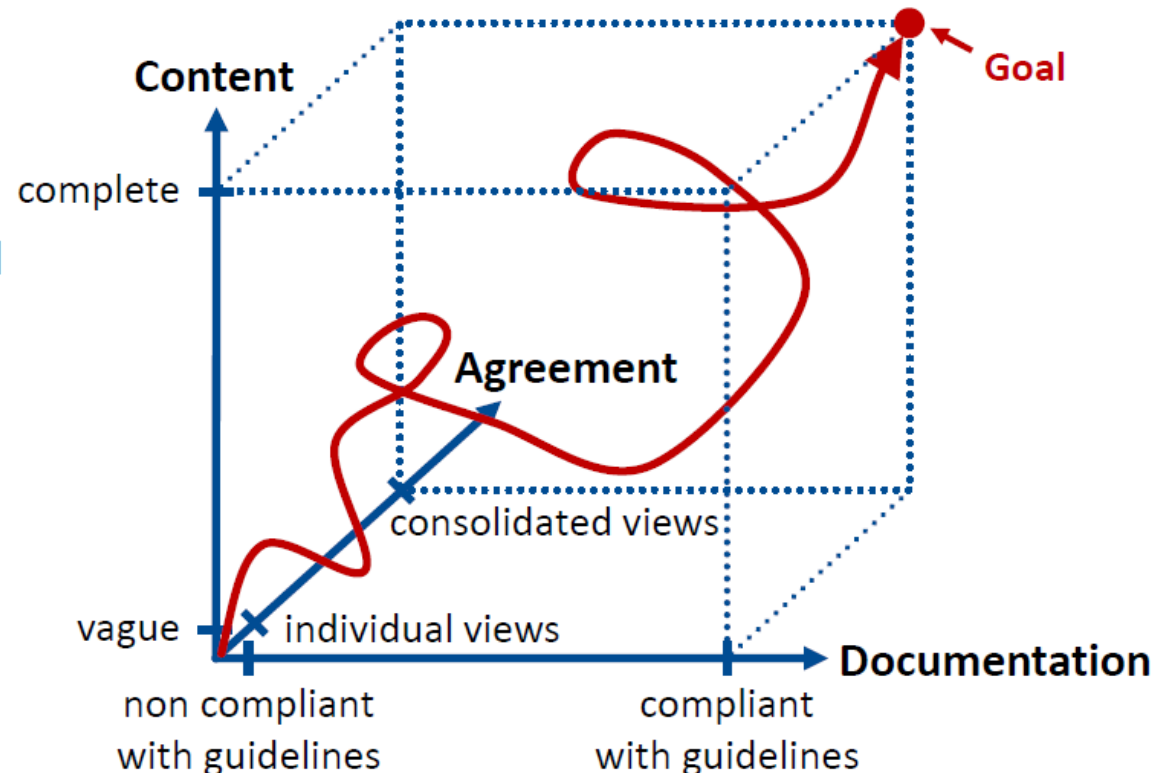


Goal of Requirements Engineering

Requirements engineering is a cooperative, iterative and incremental process which aims at ensuring that:

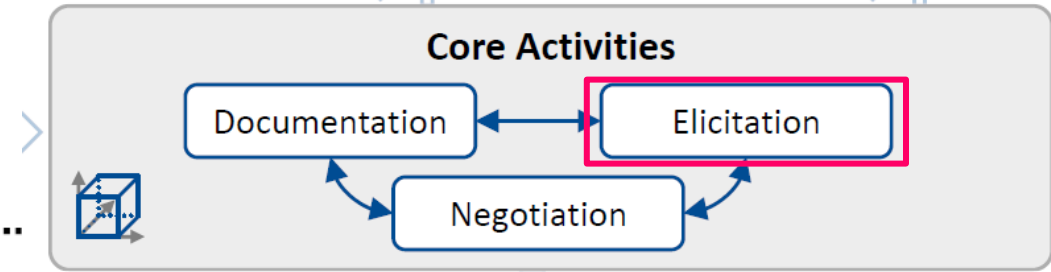
- 1) All relevant requirements are explicitly **known and understood** at the required level of detail.
- 2) A **sufficient agreement** about the system requirements is achieved between the relevant stakeholders.
- 3) All requirements are **documented and specified** in **compliance** with the relevant documentation/specification **guidelines**.

Revisit!



Based on [Pohl and Ulfat-Bunyadi 2013]

Elicitation



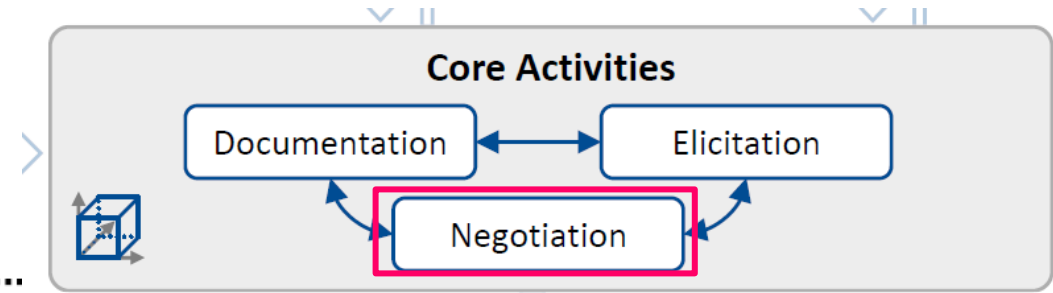
D The goal of the elicitation activity is to:

- (1) **Identify** relevant **requirements sources**.
- (2) Elicit **existing requirements** from the identified sources.
- (3) Develop **new and innovative requirements**.

- Requirements sources are **not always known** at the beginning of the process!
- Relevant requirements sources include:
 - **stakeholders** involved in the process,
 - existing **documents**,
 - existing **systems**.

Progress in content dimension!

Negotiation



D The goal of the negotiation activity is to:

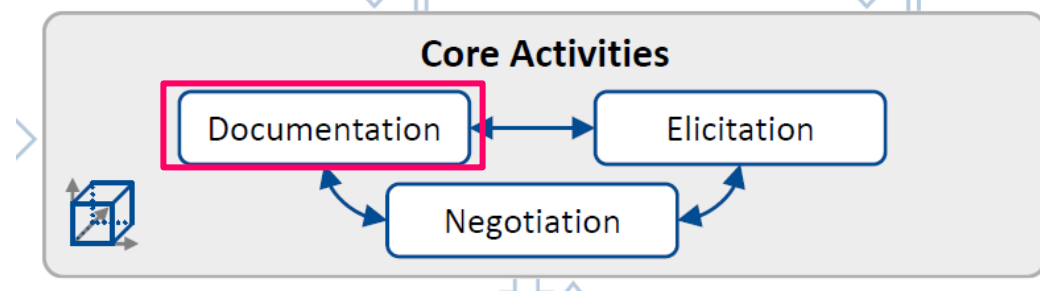
- (1) Identify conflicts.
- (2) Analyse the cause of each conflict.
- (3) Resolve the conflicts by means of appropriate strategies.
- (4) Document conflict resolution and their rationale.

- Wishes and needs of the stakeholders typically vary and can be in conflict.
- Conflicts shall be identified and should be resolved.

Progress in agreement dimension!

Three Core Activities

Documentation



D The goal of the documentation activity is to:

- (1) Document relevant requirements information according to the defined documentation guidelines.
 - (2) Specify requirements according to the defined specification guidelines.
 - (3) Choose documentation formats and notations which fit the stakeholder needs, and are defined for the project.
 - (4) Ensure consistency between different documentation formats used.
- In addition to the requirements, information about elicitation and negotiation should be documented.
 - Early in requirements engineering, information is often documented informally/unstructured and thus not compliant with the documentation and specification rules.

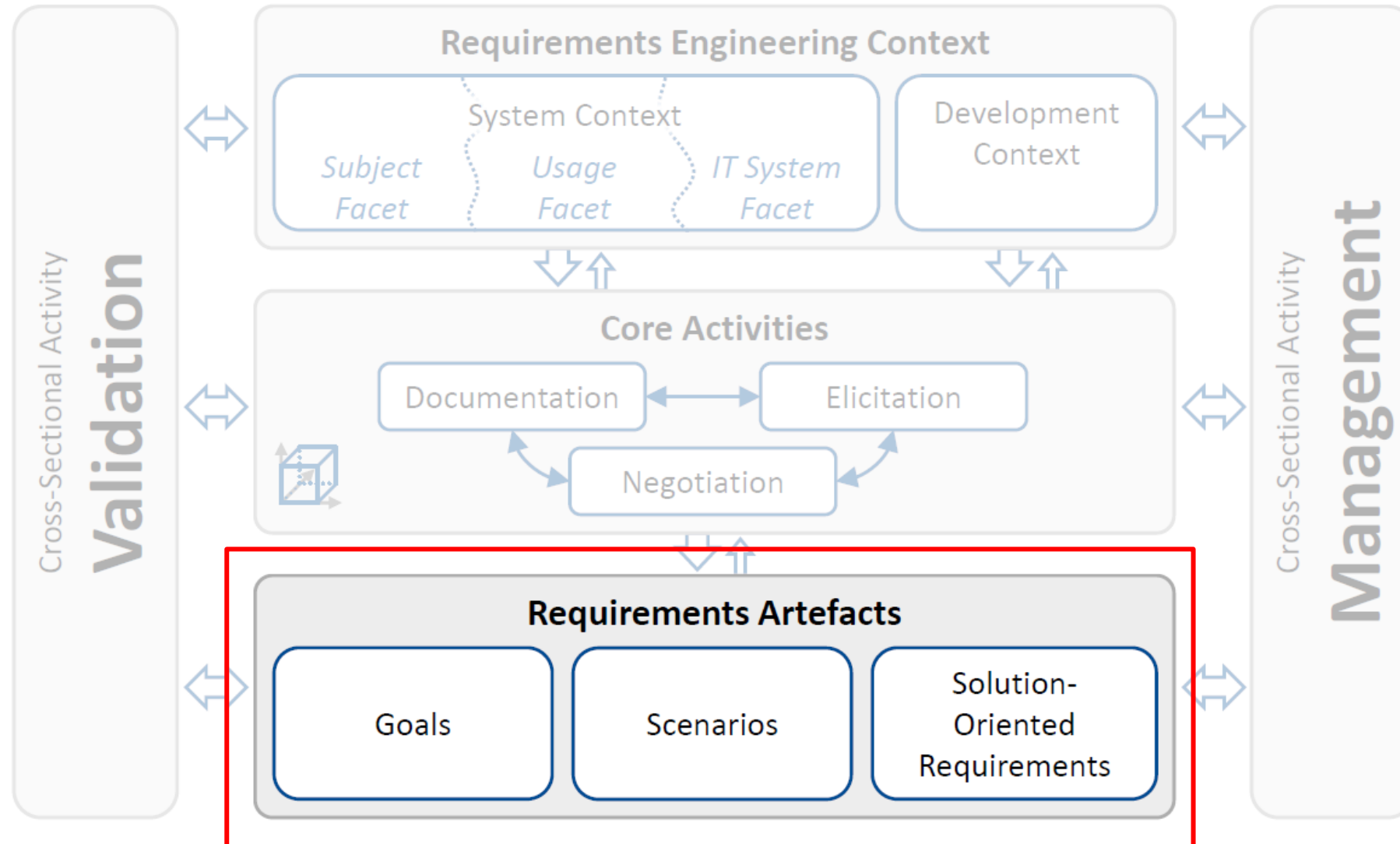
**Progress in documentation
dimension!**

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Framework for Requirements Engineering

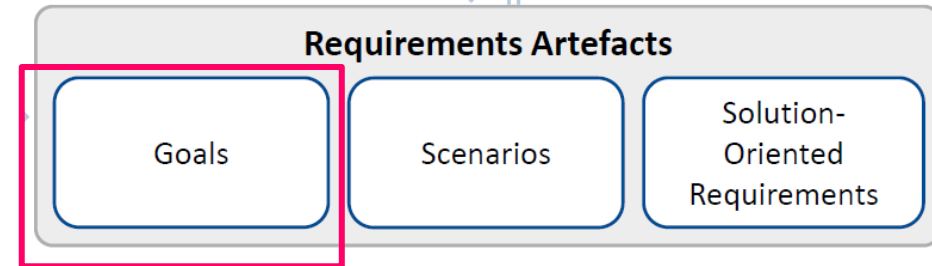


Goals (1)

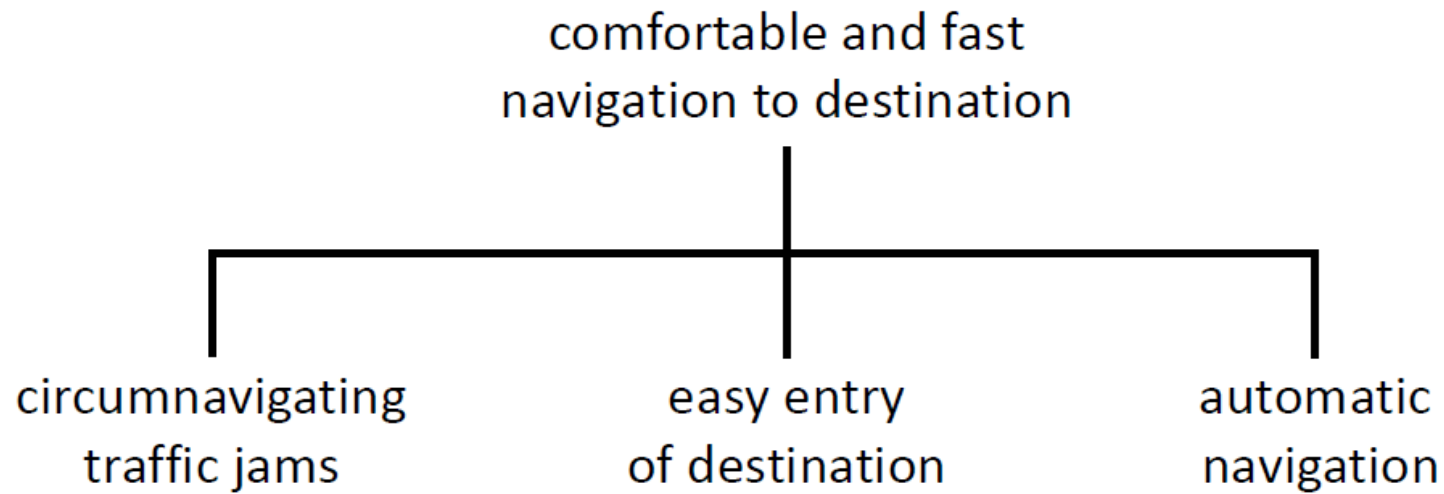
D A goal describes a high level objective of one or more stakeholders about a property of the system to be developed or the development project.

Characteristics of a goal:

- Prescriptive nature.
- Expresses stakeholders' intentions.
- Refines the system vision into more detailed objectives to be fulfilled by the system.
- Should be solution free.



Goals (2)

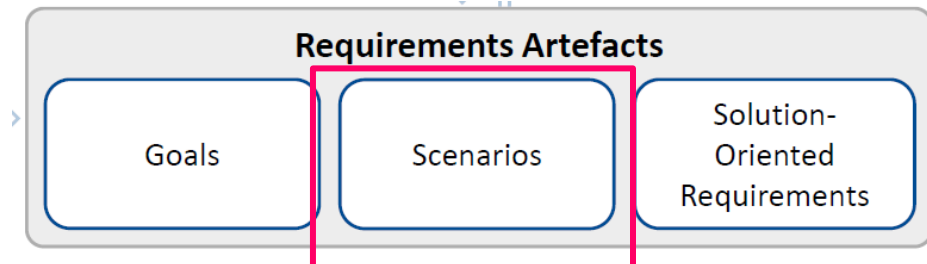


Scenarios (1)

D A scenario describes a concrete example of satisfying or failing to satisfy a goal (or set of goals).

Characteristics of a scenario:

- Documents a concrete example of system usage.
- Typically defines a sequence of interaction steps executed to satisfy (or not satisfy) a goal or set of goals.
- Puts requirements into their context.
- Increases the comprehensibility of goals.



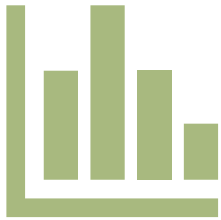
Scenarios (2)



Entry of destination

1. Driver selects the navigation to a desired destination.
2. Navigation system asks for the address of the destination.
3. Driver types in the address.
4. Navigation system checks the inserted information.

3 types of Solution-oriented Requirements



data



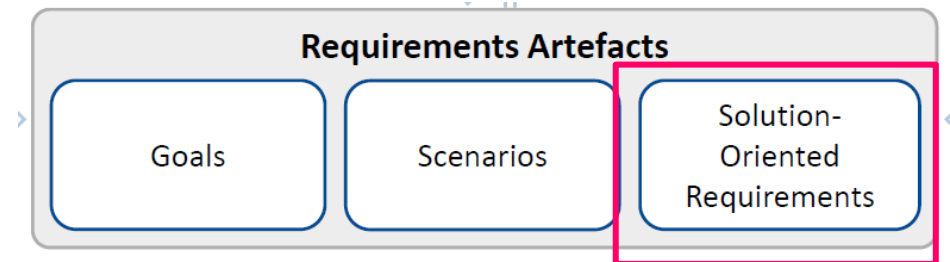
Functional



behavioral

Three Types of Solution-oriented Requirements (1)

- (1) **Data** perspective considers the static data structures. It defines data types, attributes and relationships between data types.
- (2) **Functional** perspective considers the manipulation of data by system functions. It defines the transformation of inputs by functions into outputs.
- (3) **Behavioural** perspective considers the system behaviour. It defines the reactions to external stimuli in form of permitted states, transitions and outputs.



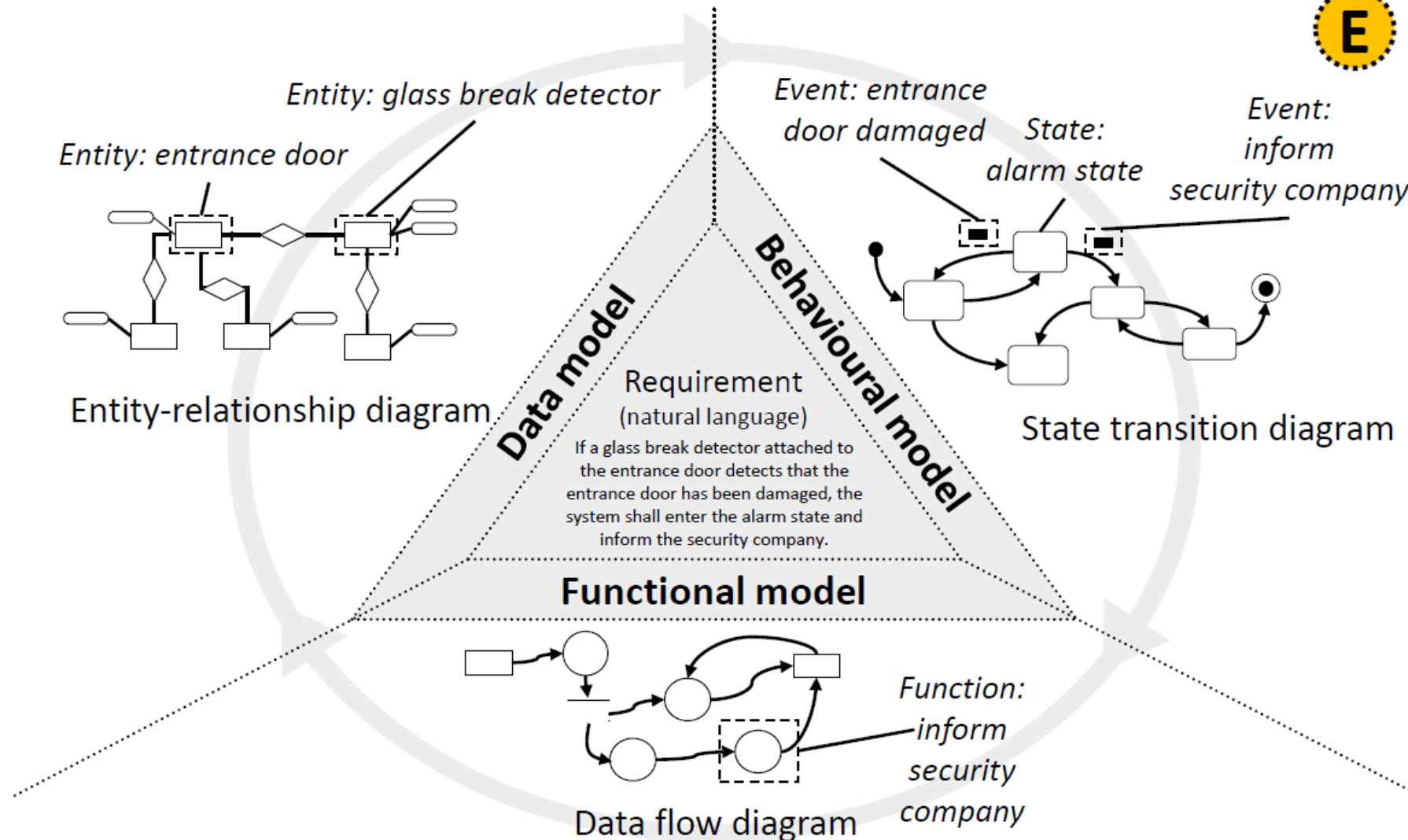
Three Types of Solution-oriented Requirements (2)

Solution-oriented requirements:

- Specify requirements at a level of detail sufficient for supporting later development activities such as design and test.
- Often imply a conceptual/logical solution.
- Should be conflict-free.
- Should be agreed on by all stakeholders.
- Should be as complete as possible.

Three Types of Solution-oriented Requirements (3)

E

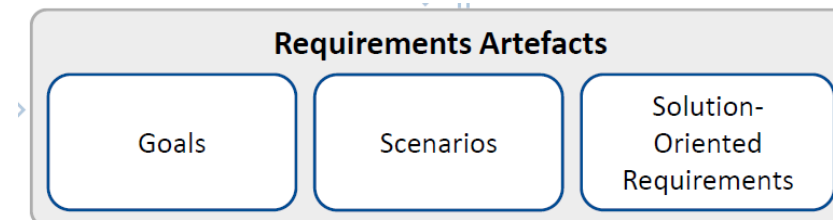


Goals, Scenarios & Solution-oriented Requirements

Goals, Scenarios and solution-oriented requirements are used complementarily during requirements engineering.

Interrelations:

- Developing goals and scenarios prior to or along with solution-oriented requirements leads to a significant improvement of the quality of the requirements.
- Scenarios put requirements into context and thus provide a basis for deriving and developing solution-oriented requirements.
- Goals and scenarios support the refinement of requirements across different layers of abstraction.



Two Categorisations



Revisit!

Types of requirements

- Functional requirements
- Quality requirements
- Constraints

Requirements artifacts

- Goals
- Scenarios
- Solution-oriented requirements

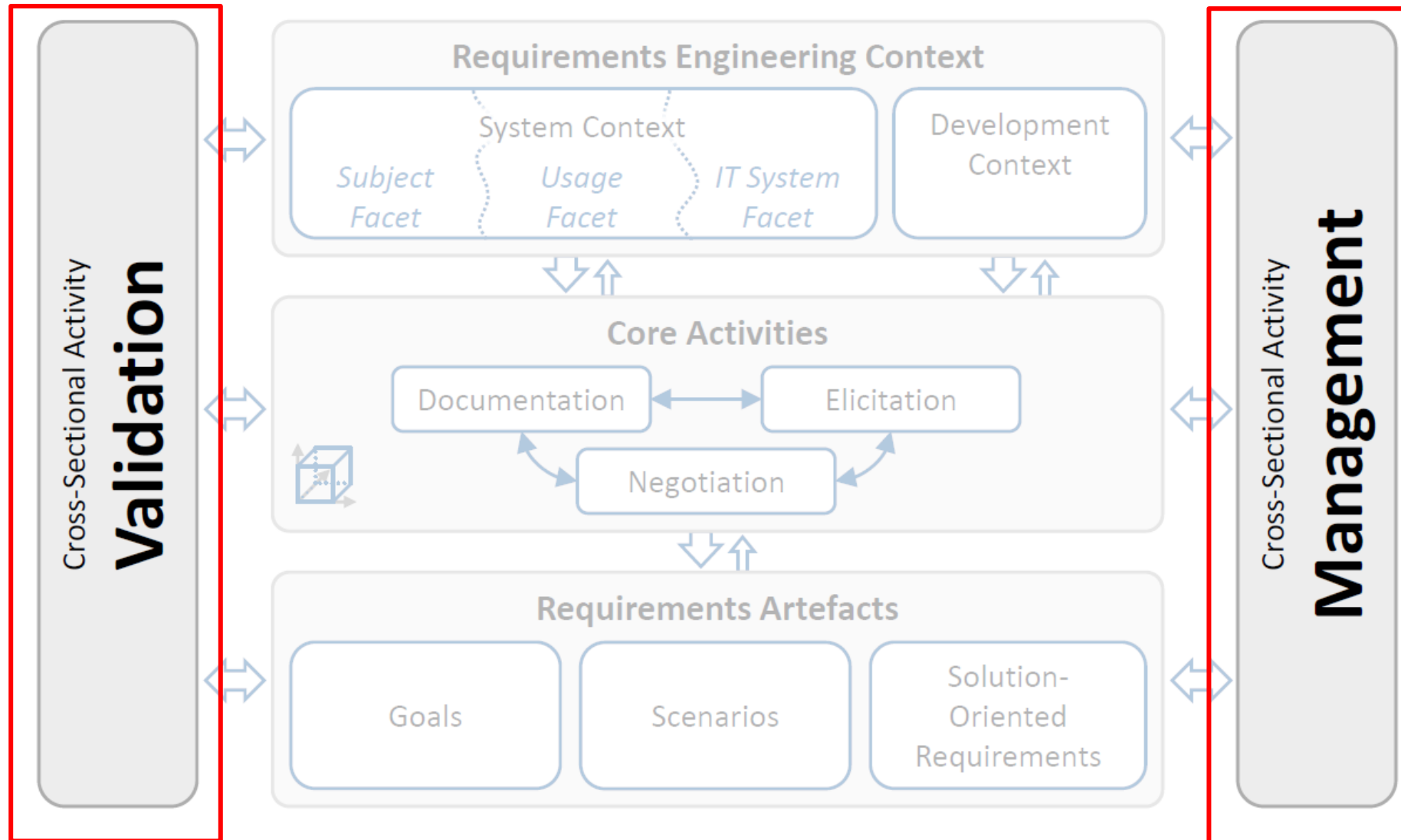
Functional and quality requirements, as well as constraints can be defined by goals, scenarios and solution-oriented requirements.

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Framework for Requirements Engineering



Three Validation Goals

(1) Validation of requirement artifacts.

Aims at detecting defects in requirements.

- Validating the artefacts with regard to the content, the documentation and the agreement dimensions.
- Checking the fulfilment of validation criteria.

(2) Validation of the core activities.

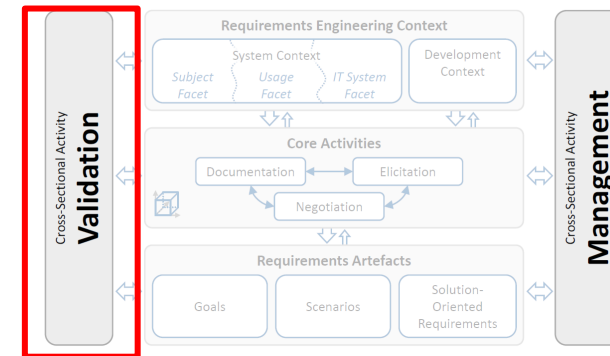
Checking the compliance of the activities performed and the process and/or activity specifications.

- Have all required steps been performed?
- Have the required stakeholders been involved in the activities?

(3) Validation of the context consideration.

Checking whether the context has been considered adequately in the activities.

- Have all relevant requirements sources been considered?
- Have the required stakeholders been involved?
- Have all parts of the requirements engineering context been sufficiently considered?



Three Management Goals

(1) Management of the requirements artifacts throughout the system lifecycle:

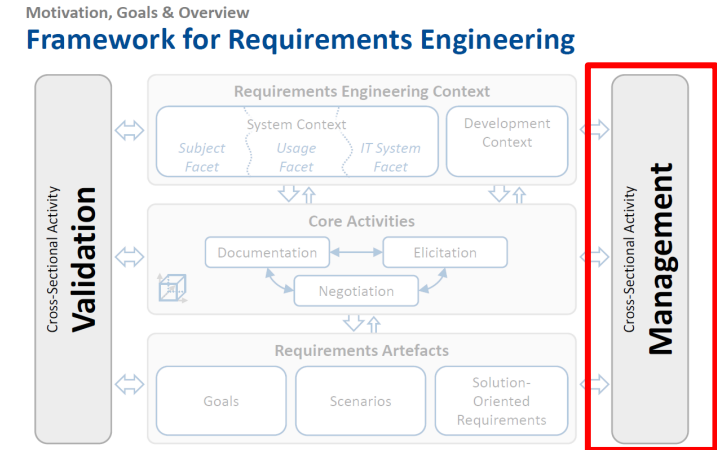
- Prioritization of requirements.
- Persistent recording.
- Configuration management.
- Change management.
- Requirements traceability.

(2) Management of the core activities:

- Ensure an efficient and effective overall RE process.
- Plan and control the execution of the RE activities.
- Alignment of the process to the current project situation.

(3) Management of the context:

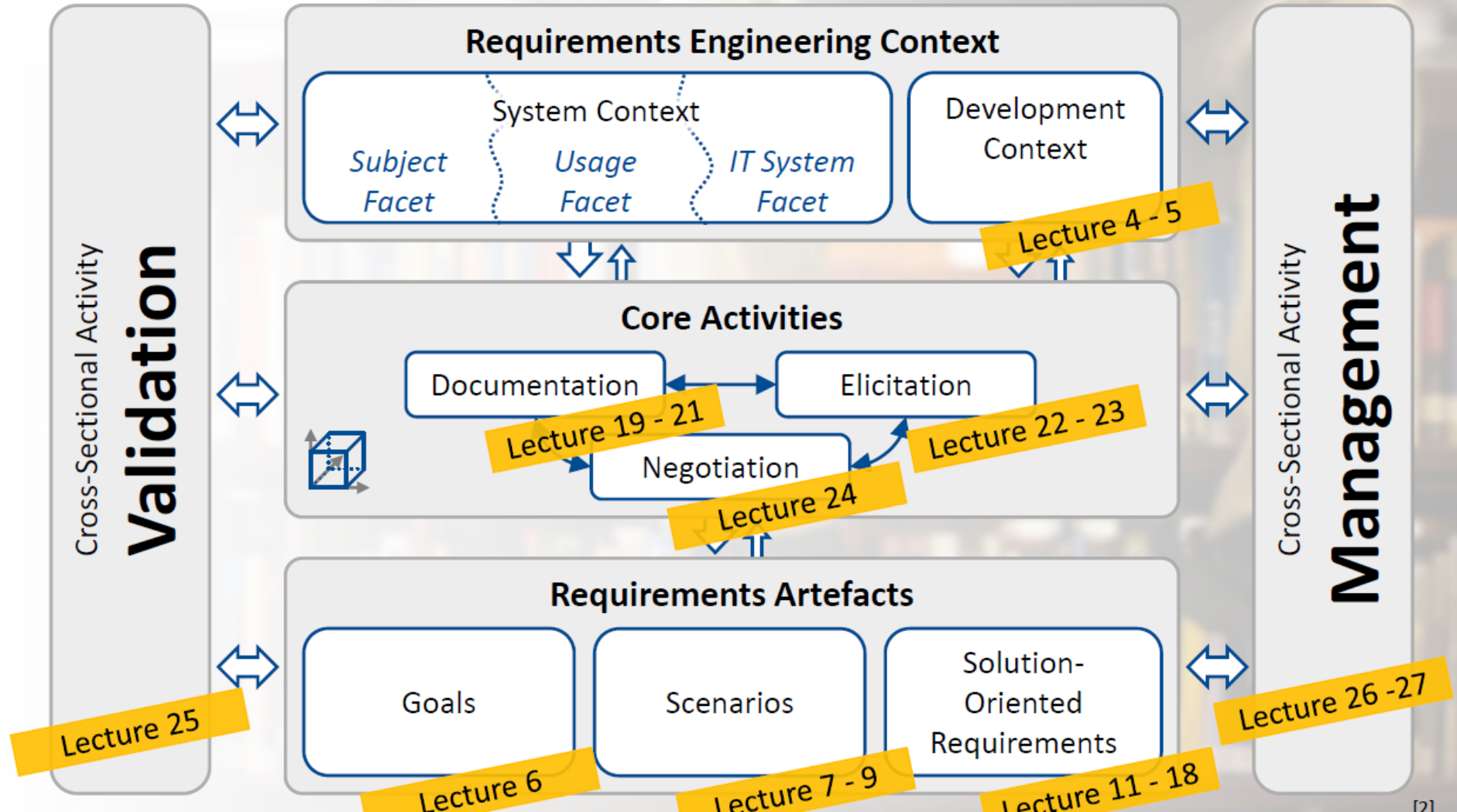
- Identifying changes in the context relevant for the system.
- Changes typically require (re-)initiating or (re-)scheduling of one or more RE activities.



Summary

- Requirements Engineering: Establishing Vision in Context.
 - Small changes in the vision might lead to huge changes in the requirements and the way the system is developed.
 - Requirements always depend on their context.
- Framework for Requirements Engineering with four building blocks:
 - (1) **Requirements Engineering Context**: structured into system context with three facets (subject, usage, IT system) and development context.
 - (2) **Three Core Activities**: documentation, elicitation and negotiation.
 - (3) **Three types of Requirements Artefacts**: Goals, scenarios and solution-oriented requirements.
 - (4) **Two Cross-Sectional Activities**: Validation and management.

A Framework for Requirements Engineering



Questions for Self-Assessment

- What are the characteristics of a vision?
- What is the purpose of the framework for requirements engineering?
- How is the RE context structured?
- Name and describe the core activities of requirements engineering.
- Name and describe the types of requirements artefacts used during requirements engineering.
- What are the goals of the cross-sectional activities (i.e., validation and management)?

Questions ??

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